Unit 5 - Managing Networks
Assignment 15: Managing a LAN Task 7&8

Introduction

Task description - *You are a junior consultant for T5 Alliance. The work you have done so far for Competition Bikes Ltd has been very well received. As they have no IT support staff of their own, they are interested in outsourcing the ongoing management to your company. First, however, you must prove your knowledge of network management.*

Task 7

(U5:P3) - Identify emerging network technologies

Task description - *Describe the following emerging technologies:*

- *Server Virtualization*
- *IPv6*
- *Cloud Computing*

Introduction

For this documentation, I will be identifying and describing the emerging network technologies of the modern day, essentially giving a brief definition on what they do and how they do it. Which I will then carry on to describing the potential impact that they bring later on the documentation, where I will be explaining and determining whether they are capable of improving the tasks.

Essentially, I will be identifying and describing Server Virtualization, IPv6 and Cloud Computing, giving a definition, the application it has on the network and how it functions within a system. Basically, explaining how each of these three new technologies carry out their tasks.

Server Virtualization

Server virtualization is basically the separation, otherwise called “partitioning”, of a physical server into smaller virtual servers. There isn’t a limit to how many partitions of virtual servers there can be within a physical server, however it is limited to the performance and other hardware capabilities that the server contains.

Essentially, server virtualization is the act of creating partitions of a server into smaller virtual servers, so that it can help to maximise the server resources. Making a single server run as many programs as multiple servers without virtualization. This division of the server is usually

done by the administrator, within a company. Who uses a software to divide the physical server into the isolated virtual environments.

An example of a popular software, which is used to divide a physical server into multiple virtual servers, is a software called VMware. Which is now not considered to be so recent, in terms of it's introduction date, however the technology has seen much wider use recently due to the improving hardware capabilities of servers and even desktop computers.

To elaborate, VMware was founded in 1990, it is basically made up of both desktop and server virtualization. It creates virtual servers by using the server/computer's RAM, CPU, GPU and other essential components such as the Hard Drive. This is so it has the capability to run the virtual machine or server itself without suffering from significant performance issues, causing the device to carry out the intended task poorly.

Usually, the resources of the server itself are hidden in server virtualization, which is referred to being masked. Meaning that accessing the physical server is hidden from the users, with the exception of the administrator.

The reason being is that the software is used to divide the physical server into multiple virtual environments, called virtual or private servers. Where the main server will be only accessible by the admins, due to the fact it isn't really necessary for the users to access it. This is in contrast to dedicating one server to a single application or task.

Overall, server virtualization is a technique in which it involves in partitioning a physical server into smaller virtual servers. What this means is that, in a single server, there are multiple copies of a "server" known as a virtual server.

These virtual servers have no difference in functionality to a full server, the only difference being is that a single physical server can have as many as the hardware can handle. Which is usually is done with the help of virtualization software, such as VMware.

Basically, server virtualization allows each server to run multiple servers, in the form of virtual servers, simultaneously. Also, in theory it is possible to run as many servers as possible, but it isn't actually possible due to the fact that the server's hardware capabilities can't allow for it to happen.

IPv6

In retrospect, an IP address is the code for identification, whether it be numbers, alphabetical symbols or both, for human readers in order to identify an individual machine connected to the internet. Specifically, the IP called Internet Protocol Version 6, specifies the format of the protocols and the addressing scheme across multiple networks which supports the use of IP.

Basically, IP supports unique addressing for computers on a network. Essentially, the way it works very similar to a house address would. Which is to identify the different types of devices that are found in a connection within multiple forms of the same device. For instance, if a user wants to send one data packet to another user on the internet they would use this protocol to find that user and send the data.

One version of this IP is the IPv6 standard, which features addresses typically 16 bytes, or 128 bits, in length. This protocol is used for the same reason as the IPv4, which is to identify a device within a network, but can have a larger amount of unique users when compared to IPv4.

However, the reason for creating an updated version of the previous protocol is because the IPv4 IP address is meant to be unique, meaning that only one device can have one allocated IP address. However, as it is only 4 bytes/32 bits in length, the use of unique IP addresses are running out for the new demand of users increasing in the modern world.

Which led to the creation of the newer protocol, which is longer in length and will not need an upgraded version of this protocol for a long period of time, due to the significantly extended identification that the hexadecimal identification brings, which is a type of alphanumeric style of identification.

To simply put it, IPv6 expands on addressing and routing capabilities when compared to IPv4. Giving it the ability to be used on a larger amount of devices, giving more devices a unique address than mathematically possible with IPv4.

Overall, IPv6 is a new concept for replacing IPv4 in order to increase the number of devices available on the network. Which can prove to be advantageous, especially in the future of technology, which I will be explaining in the next section of the document.

Essentially, IPv6 was created as an expansion to IPv4 in order to increase the amount of devices needed to connection to a network with a unique identification from the rest. This is so there is an element of future proofing the capabilities of the network, which is an essential task in order to keep everything capable of being operated in uniform order.

Cloud Computing

Cloud computing is simply the operation of performing computer tasks, using services provided over the Internet. This can range from computer resources such as virtual machines, storage, applications or basic utilities such as documents, files or images. Instead of having a built in a computer infrastructure in the same building.

Essentially, cloud computing is a type of computing where it relies on a connected network sharing computing resources, instead of having local servers or personal devices handling the applications within a system.

This type of system allows employees to work remotely on the services provided over the shared network. These services may include applications, storage systems and multiple other systems which are essential in the operation of a system, including the operating system.

Usually, companies which provide cloud services enable users to store files and applications on remote servers, and then access all the data over an internet connection. There are three types of cloud computing services which are the private cloud, public cloud and hybrid cloud. Which offer similar services, but on different types of networks.

Firstly, public cloud services are typically used for e-commerce sites such as Amazon, Azure, Microsoft and any service or application publicly accessible. Which is estimated that about 1/3 of the overall companies and organisations are moving to the public cloud, in order to adhere to the modern market and other services for customers.

Secondly, private clouds are hosted onsite at an organization for data center use. Meaning that it is used only inside a company for the employees and IT professionals found within the network. An example of companies that use and host private cloud are core business assets, such as financial applications and HR.

Lastly, enterprise clouds are used to run modern business applications for organizations of premises. Essentially, this could be a combination of both private and public accessibility, in order to achieve the work effort needed for the company to make a profit from this technology.

Overall, cloud computing is the idea of taking all the heavy lifting involved in crunching and processing data away from the device you carry around, or sit and work at, and moving that work to huge computer clusters far away in cyberspace.

Essentially, the data, work and applications are available from any device with which the user can connect to the internet, anywhere in the world. Usually, the services that are provided and used over the external network, or the internet, are paid by the organisations or the cloud customer as either a pay per use or monthly subscription.

The reason it's called cloud computing because the information being accessed is found in "the cloud" and does not require a user to be in a specific place to gain access to it. The terms “cloud” and “Internet” have become interchangeable and typically mean the same thing when technology firms reference using the “cloud” for computing.

Basically, this is a very simple to understand method of computing, but extremely complex in structure, especially on modern day technology which is much more sophisticated. It is simple to understand and complex in structure due to how it uses the services over a network, either to one user or multiple users.

Conclusion

In conclusion, each of these technologies has been created for the purpose of increasing the efficiency of the future of technology. Which is to reduce the amount of devices and hardware needed, while increasing the amount of users possible. Basically increasing the efficiency of technology.

Server virtualization is basically the creation of multiple server operating systems to run on a single physical server. The amount of physical servers available only depends on the hardware that is built on that physical server.

IPv6 is basically an enhancement of the structure of IPv4 where, instead of using numeric addresses as identification, it uses hexadecimal addresses to expand the amount of combinations. This is so it can increase the amount of nodes by increasing the amount of address combination, therefore it would be able to expand to the world wide growth of internet connected devices.

Cloud Computing is basically the gathering of services, known as cloud services, and moving them externally to the organisation's internal network. It is a service simple to use and operate but very complex in structure, due to the fact it is using different types of computer services over a network.

Basically, these technologies are being accepted and implemented into the modern technology to increase efficiency. However, it is going to be a lengthy process which may take decades, due to the fact that these are new, emerging technologies which still don't see as much usage as the previous methods.

Task 8

(U5:M1) - Describe the potential impact of emerging network technologies

Task description - *Explain how the emerging technologies listed in Instruction 7 could be implemented at Competition Bikes Ltd. For each technology, discuss the potential impact in terms of:*

- *What is new about it/how is it different from the previous way of doing things*
- *Whether or not you think they make things easier for users and IT professionals and why*

Describe;

- *What is new about server virtualization*
- *How are virtual servers different from physical servers*
- *What is new about IPv6*

- *How is IPv6 different from IPv4*
- *What is new about cloud computing*
- *How is cloud computing different from hosting your own IT systems on your own premises*

Introduction

As mentioned previously, with the identification and definition of the three emerging technologies such as Server Virtualization, Ipv6 and Cloud Computing. Now there is a clear understanding of the operating and purpose of each of the technologies, but what is new about them and why should they replace the current technology used for the similar purpose.

This is so I can prove whether the newer emerging systems is (or isn't) suitable for the company and whether it would be a good replacement for the current systems available. Essentially, in this section, I will be explaining what new features the technologies bring and how it is different from the previous way of carrying out the similar tasks.

Also, explaining whether or not I believe that they make tasks easier for the users and IT professionals, by comparing them to the previous method of these systems and comparing on them both and if the emerging technology is suitable enough to be a replacement, basically why it makes it easier.

What is new about server virtualization

Before server virtualization, only one server could have the functionality and features of a single server. This use of a server is straightforward but essential to a network, due to the fact it is where all the data in the network is stored. Which the user could recieve or send data through the network to different computers which are connected to the same network.

Now that server virtualization is available, servers can now be configured with a certain processing, memory and storage ability so that it can handle multiple virtual servers. Which would each carry out the same job as the previous single server without virtualization, meaning that the different users that use the network can access multiple servers through the connection of a single server.

As mentioned previously in the document, server virtualization is used for distributing the system data from a single server into several smaller virtual servers, which is basically dividing a physical server into smaller virtual servers so it will help to maximize the server resources.

These virtual servers can have resources such as number physical servers within a network. However, these server resources are usually masked, meaning that these resources are hidden or masked from users automatically in the name of the software is Virtual Server or Private Server.

By completely reproducing a physical network, network virtualization allows applications to run on a virtual network as if they were running on a physical network but with greater operational benefits and all the hardware independencies of virtualization, which will be mentioned.

With the introduction of Network virtualization, it has presented logical networking devices and services, such as logical ports, switches, routers, firewalls, load balancers, VPNs and many more to connected workstations, or PCs.

Overall, server virtualization is a new method in accessing a computer system through the means of a network, rather than the hard drive found on the device itself. Which has many benefits that can make operating a network easier, which will be mentioned in the next section of the document.

Essentially, with the introduction of server virtualization, it enabled the server to have the function of multiple servers, which wasn't previously available. Due to the lack of hardware capabilities of older servers and the software to partition the virtual servers in the first place. Basically, decreasing the amount of hardware and the costs while still having similar output.

How are virtual servers different from physical servers

Server virtualization has many benefits when compared to physical servers, which can make services easier for the user and IT professionals. To elaborate, there are distinct advantages in virtual servers when compared to physical servers, in the terms of making the network easier for users and IT professionals. Since it enables multiple operating systems to run on a single physical server as highly efficient virtual machines.

Server virtualization first and foremost brings greater system efficiency per server. What I mean by this is that it reduces the amount of power and maintenance needed to run a network with the installation of server virtualization. Due to the fact it reduces the amount of hardware needed to actually run the different virtual servers, rather than multiple physical servers.

Meaning that it also reduces the operating costs for each server due to the fact that less hardware needs to be purchased and maintained to keep the server in working order. So, not only does it reduce the amount of physical servers, server virtualization also reduces costs because less hardware is required so that alone saves a business money. Making it easier for the company to implement in a costly level, and for the users and IT professionals to access and maintain easier.

Also, if the server has high end hardware, it is possible that it brings an increased performance to the network applications. What I mean by this is that, due to the high end hardware on a physical server containing multiple servers. It would increase the response rate of those virtual

servers found inside the physical server, which is beneficial to the users and IT managers within a network.

Due to the fact it also means there would be much faster workload deployment. Meaning that it would increase the work rate with the employees, therefore increasing the work output and the generated income to the company as a whole.

In turn, eliminating the server sprawl and complexity. Which essentially means that the hardware in the server is being used up to the full potential, instead of being wasted. Therefore, the cost of the server is much more efficient due to the fact that more potential is being used.

Also, server virtualization brings fewer physical servers implemented, meaning that there is higher server availability within the network. Also, it lets each virtual server run its own operating system, meaning that each virtual server can also be independently rebooted of one another.

Additionally, server virtualization also conserves space through consolidation. What I mean by this is that several machines can be consolidated into one server running multiple virtual environments. Meaning that it utilizes resources to the fullest, so it can also save on operational costs, especially when using a lower number of physical servers reduces hardware maintenance as mentioned previously.

As servers can be used for VPS, or Virtual Private Server, it therefore means that companies have the ability to control monitor their network systems, instead of constantly checking the user's accessibility within a network. This essentially makes it easier to keep restricted data to a few users that only have the access to the VPS.

With Server virtualization, it is possible to move it to the cloud, as a backup. Meaning that, if the physical server was to accidently shut down or is soon to reach its life span, it could be easily replaced with another server by uploading the backup to the new server. This essentially enables a fast and cost effective method of upgrading or maintenance of servers.

However, the disadvantage to virtualization is that it should only be used for applications that are designed for small scale, medium scale for the hardware capabilities or are compatible with server virtualization in the first place. The reason that virtual server shouldn't be used for high-performance applications, is due to the fact that it will be impacting on the performance significantly, because of the overhead and complexity of the application.

However, this is becoming less of an issue due to the rapid increase of hardware capabilities and software optimisation for server virtualization. Meaning that in the near future, server virtualization will make the use of a physical server with the function of a single server completely obsolete.

Overall, with these extra features, the most noticeable potential impact this emerging network could have is that it is saving money on larger businesses. As they are not having to buy multiple servers for different purposes if they do not use all the memory on the original server.

I believe that virtual servers can be beneficial to the company, only if implemented correctly however. Since it will save the cost of the organization, since the admin will not worry about the power and cooling system on the servers, it will also save the time at administer servers, and less downtime.

For instance, with the increasingly common usage of this technology in web servers, I believe that using virtual servers is a popular way to provide low-cost Web hosting services for customers and organisations. Instead of requiring a separate computer/server for each web server, since multiple virtual servers can reside on the same computer/server.

Basically, with the concept of server virtualization, there is a high chance of applying an IT infrastructure with this technology as a way of minimizing costs, while also increasing the utilization of existing resources.

What is new about IPv6

Before IPv6, the only method of identifying devices within a network was through IPv4. Which is an internet protocol developed during the 1970's, which was used so that each computer could have a unique address on the internet. However, in the more modern day there are doubts with the usage of IPv4 and it’s capability.

Which brought the introduction of IPv6, formerly known as Internet Protocol Version 6. Which the newest version of the Internet Protocol, designed to hopefully replace the current version of IPv4. It was basically introduced in order to enhance the ability to support a much larger number of nodes in comparison to the ability of IPv4.

Essentially, Internet Protocol Version 6 is the Internet's next-generation protocol, designed to replace the current Internet Protocol Version 4 because of the increasing number of internet devices increasing in the modern world. IPv6 can also be called IPng, or Internet Protocol next generation.

To elaborate, in order to communicate over the Internet, computers and other devices must have sender and receiver addresses. These numeric addresses are known as Internet Protocol addresses, or simply referred to as IP addresses.

For example, IPv4 is a 32-bit numeric address is written in decimal as four numbers separated by periods, where each number can be zero to 255. For example, 68.89.31.226 could be an IPv4 address.

While IPv6 addresses are 128-bit IP address written in hexadecimal, which is alphanumeric, usually separated by colons. An example IPv6 address could be written as such: 3ffe:1900:4545:3:200:f8ff:fe21:67cf.

What IPv6 does is that it allows more users and devices to communicate on the Internet by using larger combinations of numbers and alphabets character to create IP addresses, which is known as hexadecimal/alphanumeric.

Which is what makes IPv6 the potential successor to Internet Protocol Version 4. Due to the fact it was specifically designed as some sort of evolutionary upgrade to the Internet Protocol and it was ensured that it will in fact be used with the older IPv4, predicted for a temporary amount of time however. So, the Internet and the number of people using it grows exponentially, so does the need for IP addresses.

Additionally, IPv6 reduces the need for Network Address Translation, a service that allows multiple clients to share a single IP address, but is not always reliable. Hence why it may need revision.

Overall, IPv6 is designed to allow the Internet to grow steadily, both in terms of the number of hosts connected and the total amount of data traffic transmitted. Meaning that it could increase the number of nodes that can have a unique, identifiable address.

IPv6 is often referred to as the "next generation" Internet standard and has been under development now since the mid-1990s. Which has been reviewed in the IETF, formally known as Internet Engineering Task Force standards committees.

Basically, the reason that IPv6 has been designed and is currently being installed worldwide, is to fulfill the need for more Internet addresses. Essentially born out of concern that the demand for IP addresses would exceed the available supply.

How is IPv6 different from IPv4

As I have mentioned previously, there are currently two version of Internet Protocol. Which is IPv4 and a newer version called IPv6. With the Ipv6 being significantly more recent to Ipv4 due to valid reasons. Meaning that IPv6 is essentially an evolutionary upgrade to the Internet Protocol. However, IPv6 will coexist with the older IPv4 for some time, due to the fact there are older devices which still do not support IPv6.

The main benefit, which has been mentioned previously, is the fact it increases the amount of unique identities thanks to its hexadecimal method of identification, instead of the original numerical identification that Ipv4 has. Which essentially makes it significantly easier to bring in more devices within a network.

However, while increasing the pool of addresses is one of the most common and initial benefit of IPv6 to the ease of users on a network, there are other important technological changes in IPv6 that will improve the IP protocol for the IT professionals found within the network, and even users.

The first difference that IPv6 has over IPv4 is that there will be no more NAT, or Network Address Translation. Meaning that it reduces the negatives that NAT brings to the network, which is the quality of the internet connectivity. So, without the remapping of the IP address, the internet speeds for the users within the network will be marginally better than with IPv4.

Also, there will be no more private address collisions, which is due to the fact that IPv6 contains the remaining information that a packet needs in order to reach its destination. Meaning that it reduces the need to integrate subnets that use the same IPv4 private address space. Due to this it is more simplified, meaning that it has more efficient routing. So, Instead of having to know every possible route, the routing tables will include routes to only those routers connected directly to them.

IPv6 also brings the ability of auto-configuration without the need of DHCP, or Dynamic Host Configuration Protocol. Meaning that it can bring easier administration for the IT professionals. Essentially, the administrator has the ability to connect the user's PCs and share files without having to contradict with the file-sharing protocols. Which makes it easier for the administrator to share files between selected users without being restricted.

Additionally, IPv6 has better multicast routing when compared to IPv4, which is essentially sending a single message to multiple recipients. Meaning that it is easier for the administrator or any other IT professional to send an important message to users within a network easier than with IPv4.

It also features a simpler header format. Which basically means that it has a simple method of transmission of data packets and even information about the file or the transmission of the file. Meaning that it is easier for the IT professional to identify what the type of packets are and what are they there for. Essentially, it provides easier management of information throughout a network.

IPv6 also brings true quality of service (QoS), also called "flow labeling" which isn't completely present in IPv4. This essentially means that there is a better throughput level, basically increasing the latency of end-to-end customers. Basically providing better connectivity to the users and IT professionals in the network.

Which is extremely beneficial in this modern era due to the gigabyte/multi gigabit networking speeds, voice over IP, streaming video and other high-demand services which are real-time. Due to the fact it essentially provides better handling of these new super-high speeds and that

has better regulation in terms of QoS. This is so that all routers will handle packets correctly, even allocating bandwidth according to priority.

In terms of security, IPv6 has built-in authentication and privacy support is another feature that comes with IPv6. This is essentially the identification of an individual within a network, usually through username and password. Essentially, this makes it easier for the administrator to determine whether a user is legitimate within a network, while also having the support of keeping the said information private to unauthorised users.

Lastly, IPv6 provides flexible options and extensions to the network. Meaning that the administrator has the ability to implement further applications or services within a network, increasing the functionality of the network to both users and IT professionals alike. Basically, it has the ability of supporting a lot more services and other system applications when compared to IPv4.

However, since the IPv4 protocol is the primary network protocol used with the internet, means that there are multiple devices that currently use IPv4 instead of IPv6, with some not having the compatibility of IPv6.

Meaning that most networks are required use the protocol version IPv4 as the standard IP that features IP addressing four bytes (32 bits) in length, meaning that it wouldn't be possible to implement all these benefits.

However, there has been a rise in the newest internet protocol IPv6, which gives the near future the ability to fully implement this new internet protocol, benefiting from the benefits and other features that this modern IP brings.

Overall, I believe that it is still too early to replace IPv4 with the implementation of IPv6 into networks. However, I do believe that IPv6 will become useful in the far future, especially when there is significantly less devices only compatible on IPv4.

Basically, due to the features that IPv6 has over IPv4, I believe that IPv6 will soon make IPv4 obsolete because it features the same functionality, but also with the benefits of a more modern network when compared to IPv4. Especially since it brings overall easier administration for the IT professionals within a company without the use of DHCP. Essentially making it easier for IT professionals to control the network easier when compared to IPv4.

What is new about cloud computing

Before cloud computing, the only method of using and accessing information was through the hard drive found within the computer itself, which would contain all the necessary applications and software tools in order to operate and carry out allocated tasks with a computer system.

With the introduction of the internet to the public, there have been developments of implementing the systems of a computer, such as the files, applications and operating systems into the internet rather than the hard disk drive within the computer. Which is what brought the concept of cloud computing.

It has been evident that in the past decade there has been major developments in the world of cloud computing. Cloud computing has the potential to let us interact with computers on a practical level that we never have before.

The main use of cloud computing would be saving and accessing important files from the cloud, streaming/downloading content and even using cloud computing to take control and interact with other devices, including the user's own device.

An example of a cloud computing service currently available is Google Drive by the Google company. This is essentially a free cloud computer service openly available to anyone with a google account, in which it allows the user to store and access any kind of document file on the space that has been allocated with any supported device, such as a PC, smartphone or tablet.

As I have mentioned previously, There are currently three different services that are available in cloud computing that bring in a new method of accessibility in the modern world. Which include a public, private and enterprise cloud. Each of the different services provide some level of security, where it either gives or denies access to public and private information within the cloud.

Firstly, the private cloud services are essentially delivered from a business's data center to internal users. Meaning that it can only be accessed within the company itself. Usually, this service offers the capability and benefits of the cloud. While still having the management, control and security available to the administrators/network managers in the company.

Secondly, the public cloud service is essentially a type of a cloud service provider, usually third-party, which delivers the cloud service over the internet. This is typically the most common use of cloud computing. Essentially, the public cloud services feature standard features to a computer system but are sold on demand either by the minute/hour or sometimes though long-term commitments.

Finally, a hybrid cloud is a combination of public cloud services over the internet and private cloud services found on the company building, with administration and some degree of automation between the two. Usually, with this service companies can run mission-critical workloads or sensitive applications on the private cloud and use the public cloud to handle workload bursts or spikes in demand.

Basically, there are different uses for each cloud type and why organizations may choose to host certain information and applications in an onsite private cloud. However, these type of

services aren't available in the storage of a HDD within a server or other form of computer system, due to the fact that it isn't connected or integrated within the cloud, or internet.

Overall, cloud computing brings in a new level of accessibility for the users that want to access their files over a wide range of devices, or in anyplace or anytime. Which it brings a completely new form of accessing and manipulating data.

Cloud computing typically has four types of services in terms of being accessed and managed. These services include on-premises, infrastructure, platform and software. Essentially, online computing as a service is split between what the user manages and what the "cloud" manages.

On premises essentially is the exclusion of any form of online cloud computing, meaning that it is the standard function of a computer system without internet access. Essentially, this is when the user has full control over the computer system and its connectivity to the network/internet.

Infrastructure as a service is essentially the management of server virtualization, servers, storage and overall network connection. Essentially, this is when the user can manage of half the services available, while the rest is already available on cloud. The other half which the user can manage includes the applications, data, runtime, middleware and the operating system.

Platform as a service is essentially the majority of management is done automatically by the cloud. Which is runtime, middleware, the operating system, server virtualization, server connectivity, storage and overall network connection. Essentially, this is when the user can manage only the applications and the data, while the rest is already available on cloud.

Lastly is software as a service, which is where the entirety of the computer system and data is on the cloud management side. Which contains the applications, the overall data, the runtime, middleware, the operating system, virtualization, connection to servers, overall storage and connection to networks. Essentially, the user doesn't have to manage anything and all the services are readily available for use on the system for them.

Basically, cloud computing gives the user the ability to access their files and information with the use of multiple devices without the need of management, instead of the one restricted to hard drive storage space, giving the ability for the user to access their information anywhere with an internet connection and a device which supports this access. Also giving the IT professionals to ability to manage the user's services within the network.

<u>How is cloud computing different from hosting your own IT systems on your own premises</u>

Cloud computing can prove to be much more beneficial when compared to the IT system's on a network's premises, especially with the use and wider availability of modern technology. To elaborate, with the average user containing more than one device available for the internet, it is

becoming more essential for the user to access the same information and files from the variety of devices.

One significant feature of cloud computing, is the storage of data. Whether it is extra storage or the access of information through the means of another device or computer system. Furthermore, there are many benefits to using cloud storage, most notable is file accessibility.

Which means that the files stored in the cloud can be accessed at any time from any place as long as you have Internet access. Making it easier for the user to access the information, no matter the location or device, as long as there is internet connectivity and a supporting device.

Another benefit with cloud computing is the cloud storage, which essentially provides organizations with off-site or remote backups of data, which reduces costs associated with disaster recovery. It also means that it gives extra storage space to organisations that require a large amount of storage, such as organisations which rely on customer account information and data such as Twitter or Facebook.

Another potential impact that cloud storage will have is that on businesses especially it allows for all documents to be saved without it taking up any space on their servers which means that the speed of the businesses systems will still be the same.

Cloud storage allows the organisation to have back up as there is no maximum storage on the internet. Therefore it is possible to have the PC backed up, just in case there is a fault in the computer system that has the ability to wipe the PC’s hard drive. This can also be handy to an organisation in the case of a workstation/server crash, causing them to lose vital information. Which they then have it backed up and can be viewed on any internet enabled device.

In terms of cloud computing, one difference that it brings is convenience. What I mean by this is that it brings in the availability for the user to access the network with multiple devices anywhere, as long as the device supports internet access and has the support to run the cloud computing service.

Essentially, cloud computing allows the user to view all of the documents from any device that has an internet connection. This means that if a user is having a meeting where one person isn't in the same locations, the users can both access the same document or file. Meaning that distance won’t affect the work output.

Another difference is scalability. Essentially, what this means is that it has the ability for the storage of the cloud to increase significantly easier than that of a hard drive. Since a hard drive needs the purchasing and installation of an extra disk in order to increase storage space. While increasing the capacity of the cloud is by paying extra for a subscription, without the need to installation.

Cloud computing can bring in low costs when compared to physical implementation of full computers. Meaning that a computer system can be drastically reduced in cost when connected through the cloud. Therefore meaning that an organisation can purchase devices with limited hardware in mass amounts, compared to a single full computer system for the same budget and the same ability to output the tasks essential to the organisation.

In the modern day, with the expansion of the internet, there is now high availability for the cloud. Essentially making purchasing, installing software and data processing equipment obsolete and time-consuming.

While cloud computing services that can get organizations up and running in a significantly shorter time. Meaning that it reduces the work time of IT professionals with the installation of devices, instead giving them the opportunity to work with other possible tasks, increasing the organisation of the network.

Also, cloud computing can offer scalable solutions and mobilization to an entire workforce. Which basically means that cloud computing increases the amount of work provided by a workforce. Due to the larger storage space and the reduced cost of devices available, which is due to the cloud containing the essentials for a computer system.

Additionally, as mentioned previously, cloud computing enables the users to access anytime and anywhere. However, the way that it is accessed depends in the different types of access methods. Examples of these methods include on-demand, broad network, resource pooling, rapid elasticity/expansion and measured service.

The first method is on-demand self-service. Which essentially means that cloud customers can sign up for, pay for and start using cloud resources very quickly on their own without help from a sales agent. The type of users that would use the would be standard employees within a business given an account in order to access the online system.

Secondly is broad network access. Which is the most simple one due to the fact that customers, such as users on a network, can access cloud services via the Internet. Typically, this is done without the need of a third party software to access. The type of user that would use this would be the customers to the organisation, whether it is to store account data or other forms of vital information.

The third method is through resource pooling. Where there are many different customers such as individuals, organizations or different departments within an organization are all using the same servers, storage or other computing resources. This will typically be done within a single building within the organisation, or multiple buildings in close proximity.

The fourth method is Rapid elasticity or expansion. Which is when cloud customers can easily scale their use of resources up or down as their needs change. Essentially, they get charged for

how much they use within a network. The type of user for this network would usually be IT professionals, since they almost always have different amount of time required in order to access these systems.

The fifth and final method of cloud computing is measured service. Which is basically when customers pay for the amount of resources they use in a given period of time rather than paying for hardware or software upfront. The type of user for this network would be organisation employees in general, standard employees and sometimes IT professionals. Since it can be adhered to the amount of work hours allocated to each of the employee over a day, week or month's time.

However, in a private cloud, this measured service usually involves some form of chargebacks. Which is where IT keeps track of how many resources different departments within an organization are using.

Essentially, cloud computing offers many more services when compared to a usual hard disk drive. However, the reason for this is because it isn't needed in a HDD, due to the fact it doesn't require the costs of another network or company in order for it to function.

With these different methods in cloud computing, the IT managers and users can choose which method is most beneficial to them in the workplace. Meaning that both the overall costs and the installation of unnecessary applications are reduced drastically.

Additionally, the cost of purchasing, maintaining and upgrading/replacing the HDD is initially much more expensive than the usage of online computing, even with a type of subscription.

On the other hand, there are some disadvantage to cloud storage is that users are limited by bandwidth. If Internet connection is slow or unstable, user might have problems accessing or sharing their files.

However, in order to access cloud computing, it requires the network which contains the software to pay for the costs of running the servers and the rest of the network. Therefore meaning that the user would usually have to pay for the type of usage, which wasn't present in the use of a hard disk drive.

Also, security is also another drawback to cloud computing. The reason for this is due to the fact the information is more vulnerable for attack, since it is stored on the internet. Which is a network which is openly available for any user, especially users that have the knowledge of infiltrating data systems.

Meaning that the information and applications stored within the cloud can be accessed by a user with the said knowledge. Giving them the opportunity to manipulate, steal or even remove the essential data and systems required for cloud computing.

Overall, I believe that cloud computing is a legitimately essential service which should be mandatory for an organisation, especially with the implementation of higher speed networks. However, it is also essential to have some form of physical storage devices in case of an emergency.

Basically, I believe that the main impact of cloud computing is the need for less physical devices and the amount of management. This is mainly seen as a positive, however there is also the issue of the loss of connection to the cloud, meaning that it isn't possible to access the services within the cloud.

Also with the benefit of reducing initial costs, by paying a monthly fee for applications instead of more expensive license fees. However, there are some cons to cloud computing as it is online, although it depends on the security within the cloud services. Due to the fact there is always the possibility that an unauthorised user can gain access to the systems and data.

Conclusion

In conclusion, in the history of technology, emerging technologies are occuring advances and innovation in various fields of technology. Which is essential due to the fact that various converging technologies are constantly being implemented in the modern day, which includes the implementation of different systems evolving towards similar goals.

This convergence can refer to previously separate technologies such as server virtualisation, IPv6 and cloud computing. Which all now share resources and interact with each other, creating new efficiencies. Also having the same purpose, to reduce the amount of physical devices/management while increasing the efficiency of networks.

Firstly, server virtualization is essentially the creation of many different virtual servers in a single package, which is a physical server. Virtual servers are different to physical servers due to the fact it only requires one server to carry out the operations of multiple servers, instead of having multiple servers for multiple operations.

Virtual servers makes operating a network easier because it reduces the amount of servers, maintenance, sprawl and complexity. Making it easier for the company in terms of cost, easier for the IT professionals in terms of maintenance/sprawl and easier for the users due to the easier access to the different server services.

Secondly, IPv6 is essentially a type of device identification within a network, used to separately identify different devices found within a network.

IPv6 is different to IPv4 in terms of how many devices each can support, with IPv6 having the ability of uniquely identifying significantly more devices compared to IPv4 due to its hexadecimal method of identification, instead of the IPv4 numerical method.

IPv6 can potentially make the functions of a network easier due to the ease of management and wider device availability within a network. Giving managers the ability to manage the network to a better standard and implement a larger variety of devices.

Lastly, cloud computing is essentially the separation of the management within a system. Which is essentially split between the user's device and the cloud service identified as on-premises, infrastructure as a service, platform as a service and software as a service.

Cloud computing is different to a physical computer within a business is that it reduces the user's management for a physical computer within a network, while also maintaining the same functionality and operation for the user.

Cloud computing makes the it easier for users and managers within a network because it gives the ability for managers to manage and manipulate the data found on user's computers through a network connection, rather than the user doing it remotely.

Basically, these emerging technologies such as server virtualisation, IPv6 and cloud computing are those technical innovations which represent progressive developments within a field for competitive advantage.

While the technologies that are being converged such as physical servers, IPv4 and physical computing represent previously distinct fields which are in some way becoming obsolete due to the emerging technology with the same interconnection and similar goals.

However, the opinion on the degree of impact, status and economic viability of several emerging and converging technologies vary, which is why there are still many opinions on this subject. Personally, I believe that it is still far too early to render the older methods obsolete, meaning that they still should be available for use until the devices can fully implement these three emerging technologies for mass usage at a reliable level.